A Modular Two-Step Protocol for Biomedical Assertion Verification: An Iteration-Stabilized, Retrieval-Grounded Fact-Checking Architecture

Abstract

The rapid integration of large language models (LLMs) into biomedical knowledge generation introduces epistemic risks associated with hallucination, unsupported extrapolation, and citation fabrication. In high-stakes domains such as oncology, the propagation of unverified claims may undermine scientific integrity, educational reliability, and clinical safety. This work presents a modular, iteration-stabilized Fact-Checking Agent for biomedical assertion verification under a Two-Step Protocol architecture.

The system combines structured assertion extraction, domain-restricted retrieval from authoritative biomedical repositories (NCBI, PubMed, NIH), a hybrid Retrieval-Augmented Generation (RAG) pipeline, multi-iteration validation, and consensus-based confidence scoring. Unlike conventional single-pass fact-checking systems, the proposed architecture incorporates a five-iteration verification mechanism to mitigate stochastic variability inherent in LLM reasoning and retrieval pipelines.

We formalize the distinction between facts and assertions, grounding the framework in scientific fallibilism and temporally contingent epistemology. The system outputs structured verdicts—Correct, Incorrect, or Flagged for Review—each traceable to evidence chains with citation metadata. Experimental evaluation demonstrates improved determinism, reduced unnecessary manual review load, and increased reproducibility relative to earlier single-pass RAG implementations.

This work contributes a scalable, modular, and epistemically transparent framework for biomedical assertion verification, advancing the responsible deployment of generative AI in scientific domains.

Introduction

Large Language Models (LLMs) such as Gemini, GPT-4, and Claude exhibit impressive fluency in generating biomedical narratives. However, their outputs frequently contain:

- Unsupported mechanistic claims
- Overgeneralized epidemiological statistics
- Fabricated or misattributed citations
- Historical inaccuracies
- Overconfident interpretations of uncertain literature

In biomedical contexts, such errors are not merely academic—they can influence educational materials, research synthesis, and clinical reasoning.

Existing automated fact-checking approaches either rely on static knowledge bases (e.g., FEVER-style datasets) or single-pass retrieval-augmented reasoning pipelines.

These architectures are susceptible to:

- Retrieval sparsity
- Query brittleness
- Stochastic LLM output variance
- Epistemic overconfidence

To address these limitations, we introduce a Two-Step Protocol Fact-Checking Agent designed specifically for domain-restricted biomedical assertion validation. The system emphasizes:

- Iterative evidence grounding
- Deterministic retrieval

- Structured reasoning constraints
- Consensus-based confidence calibration
- Explicit human-review routing

Epistemological Framework: From Facts to Assertions

A central philosophical shift in this work is the reframing of validation targets from "facts" to "assertions."

2.1 Scientific Fallibilism

Scientific knowledge is:

- Theory-laden
- Historically contingent
- Subject to revision
- Dependent on methodological standards

Thus, binary truth assignment fails to capture the dynamic structure of scientific validity.

2.2 Assertion-Centric Verification

An assertion is defined as:

A testable, evidence-referable scientific statement whose validity can be evaluated against contemporary peer-reviewed literature.

This framing acknowledges:

- Temporal validity
- Consensus variability
- Conflicting literature
- Evidence strength gradation

The Fact-Checking Agent therefore does not declare immutable truths. Instead, it evaluates whether an assertion is currently supported, contradicted, or insufficiently evidenced.

System Architecture

The system implements a modular three-layer architecture:

3.1 Extraction Layer (Stage 1)

Transforms unstructured biomedical text into structured, testable assertions.

3.2 Retrieval & Validation Layer (Stage 2)

Performs multi-iteration, domain-restricted evidence retrieval and verdict generation.

3.3 Confidence & Aggregation Layer (Stage 3)

Calculates agreement-based confidence and determines human-review routing.

Each layer is independently replaceable, ensuring LLM-agnostic and retrieval-agnostic flexibility.

Stage 1: Assertion Extraction

4.1 Design Principles

- Sentence-by-sentence evaluation
- Zero-omission mandate
- Scientific testability filtering
- Deterministic JSON output
- Search-optimized reformulation

4.2 Absolute Language Override

Assertions containing universal quantifiers (e.g., "all," "always," "exclusively") trigger deterministic override logic to prevent over-certification.

This reduces false positives and mitigates epistemic overconfidence.

Stage 2: Multi-Iteration Verification

Stage 2 replaces conventional single-pass RAG validation with a five-iteration verification protocol.

5.1 Domain-Restricted Retrieval

All retrieval is constrained to:

- NCBI
- PubMed
- NIH

This ensures authoritative evidence grounding and reduces web noise contamination.

5.2 Hybrid RAG Framework

Retrieved documents undergo:

- Content cleaning and normalization
- Metadata-aware chunking (400–600 words)
- TF-IDF vectorization
- Cosine similarity ranking

Top-k passages are passed to the LLM in structured evidence blocks.

5.3 Iterative Validation

For each assertion:

1. Retrieve evidence
2. Construct structured prompt
3. Generate JSON-only verdict
4. Record iteration status
5. Repeat 5 times

Iteration statuses include:

- SUPPORTED
- CONTRADICTED
- NO_EVIDENCE

This design mitigates stochastic instability.

Stage 3: Confidence Calibration

Confidence is defined as:

$$Confidence = \frac{\text{Agreement Count}}{5} \times 100$$

6.1 Interpretation

High confidence reflects internal consistency- not necessarily evidence strength.

Example:

5× NO_EVIDENCE → 100% confidence

(indicates stable retrieval failure, not certainty of falsity)

6.2 Human Review Routing

Assertions are routed for review if:

- Evidence is sparse
- Confidence falls below threshold
- Absolute language detected
- Ambiguity persists